

Modeling Volatility and Valuing Derivatives Under Anchoring

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Abstract

We develop a complete-markets model with volatility smiles, tractability, and intuitive appeal as an anchoring or habit-formation model. Like traditional stochastic volatility models, it is invariant to a multiplicative scaling of the stock price levels. The anchoring effect is that the volatility depends on the relative value of the current stock price compared to its past history, with an exponential weighting.

Keywords

anchoring, behavioral finance, skew, volatility modeling

1 Introduction

In the subject of quantitative finance there are few clues as to how to model the variables from first principles; how shares, commodities, interest rates, credit risk, and other instruments ought to behave. In the language of stochastic calculus this means that the functional forms of the drift and volatility in any stochastic differential equation are almost arbitrary. Take interest rates for example, what we want of our model is that rates should stay positive (even this is open to question), and neither blow up nor get absorbed. Apart from that, virtually any interest rate model is acceptable. Thus, there is rarely any peg on which to hang our modeling hat, so to speak.

There is one exception to this observation. And that exception is that there is a class of financial instruments that ought to be modeled by a process that has certain lognormal or geometric characteristics. With shares and exchange rates we are not concerned about the price level per se, rather we care about its new level compared with a previous one. It is the return that matters. Clearly this is different from interest rates or credit risk, where the level makes a big difference to our perception of the instrument. If we have \$1000 to spend, we are not concerned about the share price, we just have to buy more or fewer shares.

This suggests that a good model for a share price S would be the stochastic differential equation $dS = \mu S dt + \sigma S dX$ with σ being perhaps time dependent, but certainly

not being stock-price dependent. This model has the scaling property that if we multiply S by any constant then dS is increased by the same factor and so the return dS/S remains unchanged. Immediately this means that models such as CEV and local volatility, where the volatility depends on the level of S , are, according to our one modeling clue, unacceptable.

Rather than simply dismiss models such as CEV and local volatility, we should ask why they are popular. The answer is that they can give theoretical option values that more closely match those option values seen in the market, in particular in terms of implied volatility skews and smiles. Whether these skews and smiles ought to be an output of a theoretical model, via a process such as calibration, is open to question (Wilmott, 2009). Nevertheless it is common experience, regardless of option values and purely in terms of the behavior of a stock, that if a stock falls significantly from a previous high then its volatility does seem to increase. And this does point at a volatility that depends on the stock level S .

Is there any way that we can reconcile these two apparently conflicting requirements? Can we have a volatility that depends on S in a model that still has our scaling requirement? The answer is yes.¹

1.1 Anchoring

According to Wikipedia, anchoring is a cognitive bias that describes the common human tendency to rely too heavily on one trait or piece of information when making decisions. In the present context we use the word to refer to the natural tendency to compare the current level of a stock to some historical level or average. Of course, we may be misusing the word since there may be no bias at all, such a tendency might be perfectly rational. Yet it does fly in the face of almost all quantitative models of financial instruments.

In models of financial instruments it is extremely common, almost universal in fact, that information from the past is given less weight than 'information' from the future via option values, and implied volatilities. Note the use of inverted commas around the second 'information.' Obviously no one has a crystal ball. But there seems to be a preference for using quantitative numbers that anticipate what might happen in the future rather than learning from the past. These models even have a

mathematical name, Markov. Yet share prices are governed by human actions and emotions (perhaps not at the millisecond time scale!) and those humans do draw on the past for clues to the future.

2 The model

We are going to work with a model that has a memory. Let us introduce a new variable, A , representing averaging or, in the language of behavioral finance, anchoring (to an historical level) (see Tversky and Kahneman, 1974):

$$A = \lambda \int_{-\infty}^t e^{-\lambda(t-\tau)} S(\tau) d\tau. \quad (1)$$

Note that this is just about the simplest memory function that has tractable properties.

We now work with a volatility function $\sigma(\xi)$, a function to be determined, where

$$\xi = \frac{S}{A}. \quad (2)$$

Of course, if you multiply S everywhere by a constant then the variable ξ is unchanged. Thus the model has our scaling requirement.

Our model is thus

$$dS = \mu S dt + \sigma(\xi) S dX. \quad (3)$$

Such a volatility function can be made consistent with the previously mentioned observation that volatility increases when prices are low, but crucially this means low relative to some historical average, not simply low in an absolute sense since that would be contrary to our scaling requirement.

Note this is not the only memory function that has the required property; anything of the form

$$f\left(\int_{-\infty}^t g\left(\frac{S(t)}{S(\tau)}, t, \tau\right) d\tau\right)$$

would also be acceptable, but a general functional would be just too cumbersome for practical use.

We now explain the parameters in equations (1), (2), and (3).

2.1 μ

In principle, we can similarly model μ as a functional. This could give extremely rich dynamics to the S process. However, provided the standard technical requirements are met this is irrelevant for the valuation of derivatives, since our model is clearly complete and so allows dynamic hedging and thus risk-neutral valuation. When we come to determine the parameters in our model from historical time series (not calibration), we will see whether the simplest μ being a constant is sufficient to represent the statistics of the underlying.

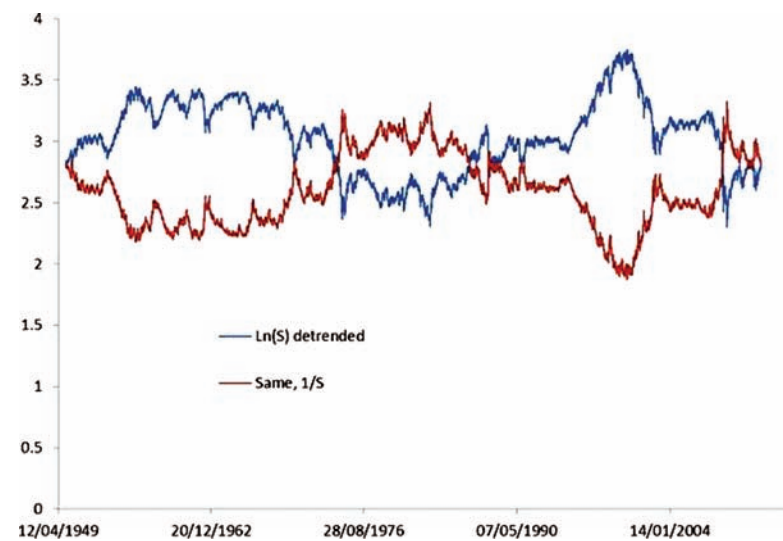
2.2 λ

The parameter λ is a measure of the extent of memory, the larger it is the shorter the span of memory. We would guess that in the financial markets (relevant) human memory would be measured in months or a small number of years, not in weeks or decades. And by 'memory' here we really mean that part of a time series that is deemed relevant to the stock's current performance, that is allows for 'forgiveness' as it were.

2.3 σ

For a simple stock we would expect the sigma function to be larger for small values of ξ than for large values of ξ . (And this may result in a theoretical negative skew.) The function $\sigma(\xi)$, being a power of ξ , might be appealing since it mimics the CEV model (although ours is more sensible, having the desirable scaling property). We will discuss this case later, where we will see that it results in undesirable properties for ξ .

Figure 1: The time series for detrended log SPX from 1950 until March 2012.



There may possibly be a minimum of $\sigma(\xi)$ if mid-values of ξ are thought of as being more stable than extreme values. (And this in turn may be seen in a resulting theoretical smile.)

We shall do most of the analytical development for general $\sigma(\xi)$ and then use a simple sigmoidal function when we discuss explicit examples.

2.4 Qualitative observation on share prices generally

Figure 1 shows two lines, one being the logarithm of Standard and Poor's 500 index (SPX) from 1950 until March 2012, detrended, and the second being the inverse of the index, also detrended and shifted vertically. Essentially the latter is just the mirror image of the former. If we were dealing with simple, lognormal random walks then the choppy (that is Brownian Motion) nature of these two lines would be qualitatively similar. But there is one aspect of the real financial time series that is different from its mirror image: the artificial data, the mirror image, has peaks and troughs like a sea wave, the peaks being pointy and the troughs being rounded. With the real data this picture is upside down. The real data has pointed troughs and rounded crests. This is suggestive of higher volatility for lower relative share prices; relative since we have detrended the series in this picture. Again, this is consistent with our model.

3 Option valuation equation

Since

$$dA = \lambda(S - A) dt \quad (4)$$

the equation governing the value of anything but strongly path-dependent options is obviously²

$$V_t + \frac{1}{2}\sigma^2\left(\frac{S}{A}\right)S^2V_{SS} + rSV_S - rV + \lambda(S - A)V_A = 0. \quad (5)$$

This will have the usual final condition, at time $t = T$ the expiration, representing the payoff. The direction of convection makes boundary conditions in A unnecessary.

4 Observations on ξ

By examining the stochastic differential equation for ξ in isolation we can tell a great deal about the behavior of this model, and get some clues as to how to approach the determination of a functional form for σ and for determining a value for λ .

From Equations (1), (2), and (3) we have

$$d\xi = (\mu + \lambda - \lambda\xi) \xi dt + \sigma(\xi)\xi dX.$$

We shall find it very useful to work with the Fokker–Planck Equation to relate the density function for ξ to the volatility function $\sigma(\xi)$. In the steady state (assuming it exists) we can relate these via the ordinary differential equation

$$\frac{1}{2} \frac{d^2}{d\xi^2} (\sigma(\xi)^2 \xi^2 P_\infty(\xi)) = \frac{d}{d\xi} (D(\xi) \xi P_\infty(\xi)), \quad (6)$$

with $P_\infty(\xi)$ being the steady-state distribution and $D(\xi) = \mu + \lambda - \lambda\xi$. Given any two out of the three functions $\sigma(\xi)$, μ , and $P_\infty(\xi)$, we can use this equation to determine the third. That is, assuming the model is correct.

From historical data for S , and hence for A , assuming that we know λ , we can determine both $\sigma(\xi)$ and $P_\infty(\xi)$. (The former by differencing the ξ time series as explained in Chapter 36 of Wilmott, 2006 and the latter by looking at the distribution of the time series data, as explained in the same chapter. These ideas were first introduced in Apabhay *et al.*, 1995.) And since we are assuming μ to be a constant, again easily determined from the time series if it assumed constant, Equation (6) can be used to check whether the model is internally consistent and robust.

5 Determining $\sigma(\xi)$ from the time series

As already mentioned, we take daily prices for the Standard and Poor's 500 index from 1950 until March 2012. First we estimate μ , assuming it to be constant. This is not done in a particularly sophisticated way, just treating the asset price as the constant volatility model. It is found to be 0.0835. Next we wish to analyze the data for ξ . Choose a λ that seems plausible – we shall shortly be performing an optimization that properly determines its value – and calculate a time series for A , using Equation (4) and a starting guess for A . Thus we have a time series for ξ . See Figure 2.

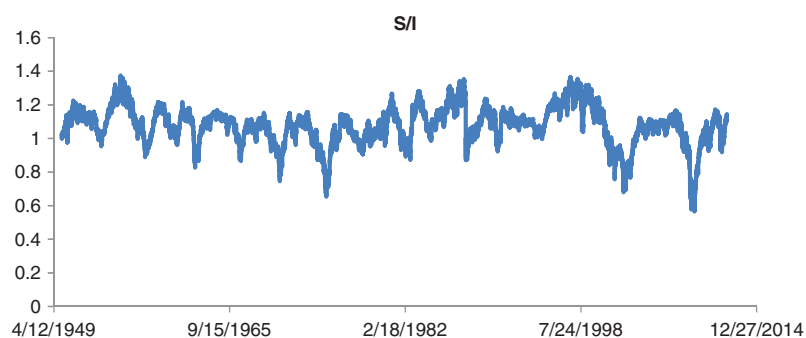
Difference this to get a time series $d\xi$, which is virtually equivalent to a time series for $\sigma(\xi)\xi dX$ since the time between data points is so small. From this we can estimate the function $\sigma(\xi)$ as explained in Wilmott (2006). The results are shown in Figure 3. (Note that this shows the $\sigma(\xi)$ after the (still to be described) optimization has been done, so in effect this is our model.)

This already shows the realistic, and expected, behavior in which the volatility is large for smaller ξ . Because we wish to manipulate this function we need to first fit a relatively simple function to it. We chose the sigmoidal function

$$a + \frac{b - a}{1 + e^{-\alpha(\ln(\xi) - d)}}.$$

The reason for this choice is the behavior for small and large ξ , where it asymptotes to two different volatility values. Figure 4 shows the original data and the fitted curve. (Again, this fitted curve is the one that is optimal as discussed shortly.)

Figure 2: The time series for ξ , SPX from 1950 until March 2012.



6 Determining $P_\infty(\xi)$ from the time series

By finding the probability density function for the data in Figure 1, we find the function $P_\infty(\xi)$. See Figure 5. However, from equation (6) we have a relationship between the $\sigma(\xi)$ we have just found and the $P_\infty(\xi)$ we have also just found. So we could for example take the found $\sigma(\xi)$ and use it to determine $P_\infty(\xi)$. By integrating (6) we have

$$P_\infty(\xi) = \frac{1}{\xi^2 \sigma^2} \exp \left(2 \int^\xi \frac{\mu + \lambda - \lambda s}{s \sigma(s)^2} ds + C \right). \quad (7)$$

Figure 3: $\sigma(\xi)$ from time series data.

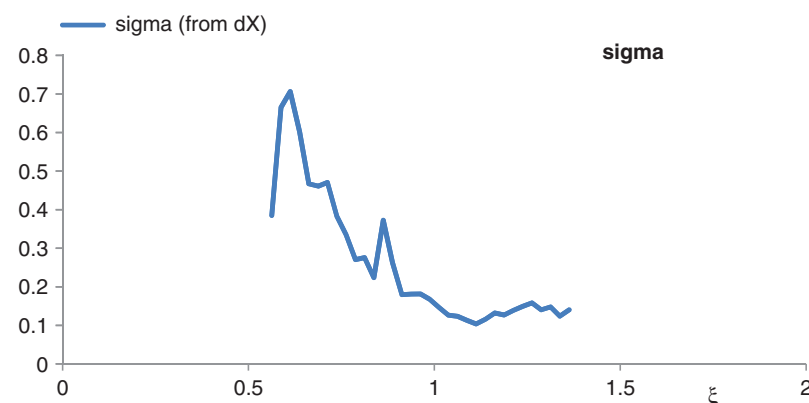


Figure 4: A sigmoidal function fitted to the data.

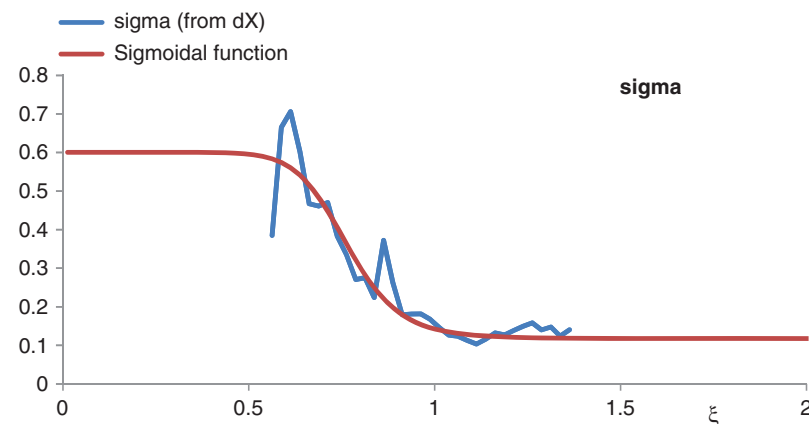


Figure 5: The steady-state PDF for ξ from time series data.

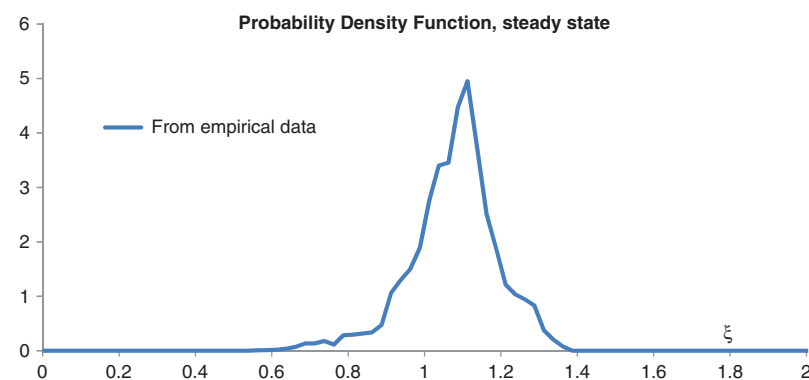
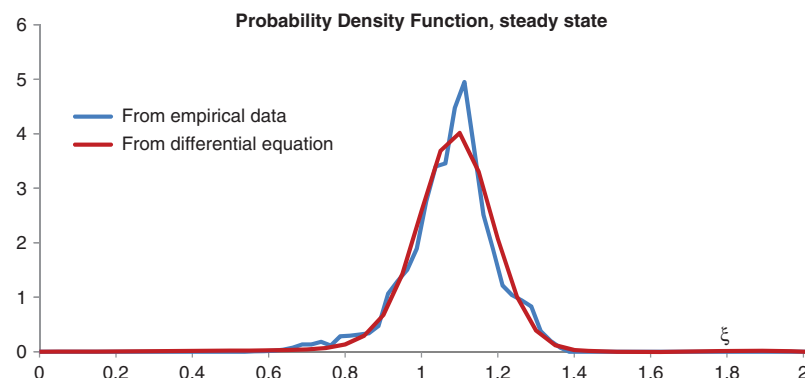


Figure 6: Best fit integrated steady-state PDF and empirical PDF.



This can be done numerically. The constant of integration C is chosen to ensure that the integral of the PDF over zero to infinity is one.³

Let us recap. We have picked a plausible value for λ , which is by no means necessarily correct. This gives us a time series for ξ and hence a (fitted sigmoidal) function for $\sigma(\xi)$ and a steady-state PDF. The latter two are also related via equation (7). There are five parameters in our model, λ and four (a, b, c, d) for the sigmoidal $\sigma(\xi)$ function. We also have the parameter μ estimated directly from the S time series.

7 The optimization

We choose λ and a, b, c, d to minimize a combined, weighted error by finding the best fits for both:

1. the sigmoidal function to the empirical σ function and
2. the steady-state PDF obtained from integrating the sigmoidal function to the empirical mean and standard deviation of the steady-state distribution.

The results of the best-fit sigmoidal function have already been shown in Figure 4. The best-fit steady-state PDF is shown in Figure 6.

The best-fit parameters are

$$a = 0.60, \quad b = 0.12, \quad c = 10.82, \quad d = -0.27 \quad \text{and} \quad \lambda = 0.90.$$

The last of these confirms a typical memory of the order of 1 year, as anticipated.

7.1 A note on the goodness of fit

We can fit the σ function as accurately as we like, by choosing suitably flexible functions with sufficient degrees of freedom. But from that function follows the steady-state distribution P_∞ . There is no guarantee that this distribution should be anywhere near matching the empirically calculated distribution given that we have allowed no freedom whatsoever in the parameter μ . Had we given ourselves the freedom to choose μ to be a function of ξ or something even more general, then such a good fit would probably have been possible. But this was not guaranteed with the simple choice of μ as a constant, and especially not with μ being the constant given directly by the time series data. Considering this, the match of the two functions to the data is remarkable.

As a test of the robustness of the model we input different values for μ and performed the same optimization. The fits were in no case anywhere near as impressive as when we used the historical value for μ .

8 Numerical results for option values

We report results from two numerical solvers for the option valuation PDE. Both use Mathematica's NDSolve.⁴ The state space for (S, A) is the (unbounded) positive first quadrant. For the first solver, we map this to the unit square (X, Y) using the transformations $X = S/(S + S_0)$ and $Y = A/(A + A_0)$, where (S_0, A_0) are the initial 'hot spot' values. With that, let $V(S, A, t) = Ke^{-r\tau} h(X, Y, \tau)$, where $\tau = T - t$ and K is the option strike price. Then equation (5) becomes $h_\tau = Ah_{XX} + Bh_X + Ch_Y$, with new (variable) PDE coefficients that are straightforward to develop. As noted earlier, boundary conditions are unnecessary at $Y = 0$ and $Y = 1$. Even if you are not a Mathematica user, our solver call for a put option should be fairly readable. It simply repeats the PDE, sets the initial/boundary conditions, and then specifies the method:

```
NDSolve[{D[h[X, Y, t], t] == A[X, Y] D[h[X, Y, t], {X, 2}]
+ B[X, Y] D[h[X, Y, t], X] + C[X, Y] D[h[X, Y, t], Y],
h[X, Y, 0] == Max[1 - (S0/K) X / (1 - X), 0], h[Xmin, Y, t] == 1,
h[Xmax, Y, t] == 0}, h, {X, Xmin, Xmax}, {Y, Ymin, Ymax}, {t, T, T},
Method -> {"MethodOfLines", "SpatialDiscretization" ->
{"TensorProductGrid", "Coordinates" -> {xgrid, ygrid}}}]
```

The (X, Y) spatial grid here is a uniform discretization in both directions: $\{i/N, j/N\}$, $(i, j = 1, 2, \dots, N-1)$, N an integer. Table 1 shows results on the implied volatility surface for various strikes with $S_0 = A_0 = 100$. The associated put option values and parameter set are found in Table 2. Note that the parameter set is different from the earlier example. The results are based on spatial grids of size 250×250 , except for $T = 100$, which uses a mesh of size 200×200 . Entries for $T = 0$ come from a separate theoretical analysis of the model's asymptotics, as discussed below.

As a second test of the numerical results we discuss how to approximate the solution of the PDE (5) using the finite difference method (FDM) and subsequent

Table 1: Anchoring model: Black-Scholes implied volatility (%)

T	K=60	K=80	K=100	K=120	K=140
0	29.46	21.17	17.13	15.99	15.61
0.25	29.03	21.15	17.18	16.07	15.76
1	27.78	21.07	17.50	16.29	15.78
3	25.68	20.91	18.15	16.85	16.22
10	24.04	21.62	19.92	18.77	17.97
100	25.60	25.16	24.82	24.54	24.31

Table 2: European-style option prices for Table 1 (p=put, c=call)

T		K=60	K=80	K=100	K=120	K=140
0.25	p	0.0004298	0.04264	2.832	18.593	38.296
	c	40.731	21.017	4.049	0.0540	0.00003977
1	p	0.1700	0.8486	4.681	16.280	33.551
	c	43.039	24.674	9.463	2.019	0.2460
3	p	0.9670	2.379	6.035	13.695	25.444
	c	49.169	33.315	19.706	10.099	4.583
10	p	1.928	3.534	5.988	9.558	14.397
	c	65.171	54.524	44.725	36.043	28.629
100	p	0.05594	0.08064	0.1073	0.1358	0.1658
	c	99.609	99.485	99.363	99.242	99.123

Parameter set: $a = 0.6, b = 0.15, c = 10, d = -0.3, r = 0.049, \lambda = 0.5$.

implementation in C++. The PDE (5) is defined on the positive quarter-plane in S and A . In order to reduce it to a problem on a bounded domain, we can choose between the popular *domain truncation* and *domain transformation* techniques. We choose the latter approach, as already introduced in this section. We then transform equation (5) in the variables (S, A) to a convection–diffusion–reaction PDE in the variables (X, Y) . In other words, we solve a transformed PDE on the unit square. In order to specify the full problem we define an initial condition (in this case the pay-off function) and boundary conditions. For the X (transformed underlying) variable we prescribe well-known boundary conditions for call and put options while no boundary conditions are needed for the transformed anchoring variable Y , as already mentioned.

In order to approximate the solution of the transformed PDE we have chosen to use the *Alternating Direction Explicit* (ADE) method (for the original source see Saul'yev, 1964 and for applications to computational finance see Duffy, 2009). There are two main ADE variants: first, the original Saul'yev scheme that is unconditionally stable and explicit and second, the Barakat–Clark (1966) scheme that is also unconditionally stable and explicit but more accurate than the Saul'yev scheme. The main advantages of the ADE method are: it is fast, easy to program, and can be applied to a wide range of linear and nonlinear time-dependent PDEs (some examples are discussed in Pealat and Duffy, 2011, and Duffy and Germani, 2013). When applied to the problems in the current work we found excellent agreement between the results produced by ADE and those produced by Mathematica. We note that ADE uses a constant mesh size in time while NDSolve uses adaptive time-stepping.

We discuss the accuracy and performance of the Saul'yev scheme, in particular as applied to put option pricing and results in Table 2. This scheme has a truncation error $O(\Delta t^2, h^2, (\frac{\Delta t}{h})^2)$ and we note the presence of the inconsistent term $(\frac{\Delta t}{h})^2$ that affects accuracy, where Δt is the step-size in time and h is the step-size in space. We can thus take a large number of steps in order to reproduce the results in Table 2. In general, we can get two-digit accuracy behind the decimal point using 200 mesh points in each direction, for example. We can improve the accuracy by using the Barakat–Clark scheme that has a truncation error $O(\Delta t^2, h^2)$ (note that there is no inconsistent term in this case). This scheme is discussed in Duffy (2009).

In order to compute the implied volatility values in Table 1 we use the Black–Scholes formula for calls and puts using the option values that we found using ADE. There are many methods to solve this nonlinear equation and in this case we used the *fixed-point method* in combination with Aitken's delta-squared process to accelerate convergence. It is worth noting that the fixed-point method converges for any initial guess because the corresponding *iteration function* is a *contraction* (Haaser and Sullivan, 1991). This is in contrast to the Newton–Raphson and bisection methods (for example), where it can sometimes be an issue to find good starting values for these iterative methods.

Finally, the ADE method is very efficient and does not need matrix solvers. In fact, most of the computational overhead lies in computing the sigmoidal function that we introduced in Section 5. This involves calling the exponential and logarithm functions, which are computationally expensive. We can then consider faster approximations to these functions or pre-computing lookup tables of the sigmoidal function. For an introduction to the finite difference method in finance, see Duffy (2006).

9 Dimensional reduction: pricing by Fourier transform

We introduce an alternative which reduces the PDE problem from two spatial factors to one factor. Apart from its application to general time numerics, we need this development for our discussion of model asymptotics.

Consider general bivariate stochastic volatility models of the form

$$dS = (r - q) S dt + \tilde{\sigma}(y) S dW^{(1)}$$

$$dy = b(y) dt + a(y) dW^{(2)},$$

where $dW^{(1)} dW^{(2)} = \rho(y) dt$. For example, the Heston model and many others fit, with $\tilde{\sigma}(y) = \sqrt{y}$, so that y is the variance rate. Our anchoring model fits with $y = \xi = S/A$, $\tilde{\sigma}(y) = \sigma(\xi)$, and $\rho = 1$, a degeneracy which is not a problem. Note that $\tilde{\sigma}(\cdot)$ is a general function, while $\sigma(\cdot)$ is the sigmoidal function. For simplicity, assume y is positive.

In traditional stochastic volatility models, for example (Heston, etc.), S_t transformations do not affect y_t , and one has scaling: invariance under $S_t \rightarrow k S_t$. Under the anchoring model, you need to transform the entire historical series: $\{S_t\} \rightarrow \{k S_t\}$. This is translational invariance for $\{x_t\} = \{\log S_t\}$. Under both types of scaling one has this key property: the marginal characteristic function $\Phi = E[e^{iz(x_t - x_0)} | x_0, y_0]$ is conditionally independent of x_0 , the conditioning being on y_0 .⁵ Thus, $\Phi = \Phi(t, y_0, z)$. This leads to the Fourier pricing formula for call options:

$$C(T, S_0, y_0; K) = S_0 e^{-qT} - K e^{-rT} \int_{\Im z \in (0,1)} e^{-izm} \frac{H(T, y_0, z)}{z^2 - iz} \frac{dz}{2\pi}. \quad (8)$$

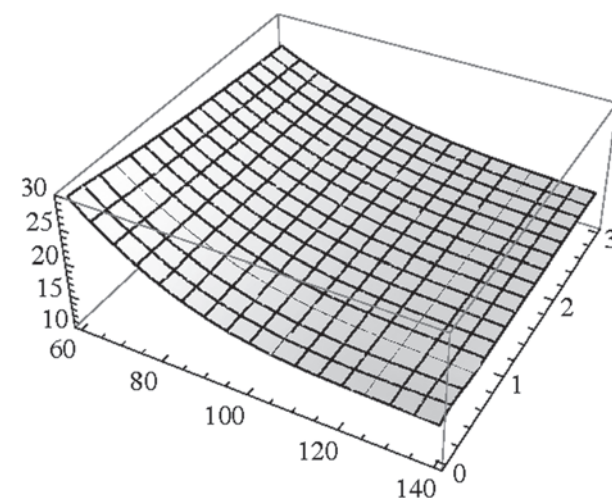
Here, $m \equiv \log\{S_0 e^{-qT}/K e^{-rT}\}$ is a natural measure of ‘moneyness’ and $H(T, y_0, z) \equiv e^{iz(r-q)T} \Phi(T, y_0, -z)$. The integral notation means: integrate along a line parallel to the real z -axis, such that the imaginary part $\Im z \in (0, 1)$. We had some discretion about where to integrate. We chose $\Im z \in (0, 1)$ because that condition coincides with a strip of analyticity of the integrand whenever the process has both a bounded norm and a bounded expected stock price.⁶ It has been shown in Lewis (2000) that $H(\tau, y, z)$ is a solution to the PDE problem

$$H_\tau = \frac{1}{2} a^2(y) H_{yy} + [b(y) - iz\rho(y)\tilde{\sigma}(y)a(y)] H_y - c(z)\tilde{\sigma}^2(y)H, \quad (9)$$

where $c(z) = (z^2 - iz)/2$. The solution we want has the initial condition $H(0, y; z) = 1$. For our anchoring model (and many others), no boundary conditions are needed at $y = 0$ or $y = \infty$. For anchoring, $H = H(t, \xi; z)$ and (9) becomes

$$H_\tau = \frac{1}{2} \sigma^2(\xi) \xi^2 H_{\xi\xi} + [(\lambda + r - \lambda \xi) - iz\sigma^2(\xi)] \xi H_\xi - c(z)\sigma^2(\xi) H. \quad (10)$$

Figure 7: Anchoring model implied volatility vs. strike and maturity. The strikes range from 60 to 140, and the times range from 0 to 3 years. The numerical method uses equation (8) with a numerical PDE solver to obtain the integrand.



As promised, we have achieved a dimensional reduction – at the expense of requiring complex numbers and an integration. The recipe here: write a one-dimensional solver for (10) and feed the solutions to (8), integrating along $z = i/2 + u$, $u \in (0, \infty)$. This Fourier method is used to produce the implied volatility surface shown in Figure 7.

One check: consider the case where the sigmoidal function is degenerate, so that $\sigma^2(\xi) = \sigma_0^2$, a constant. Then, the solution to (10), subject to $H(0, \xi; z) = 1$, is $H(\tau, \xi; z) = e^{-\alpha(z)\sigma_0^2 \tau}$. Note that this has no ξ -dependence. As shown in Appendix 2.1 of Lewis (2000), and as expected, (8) then yields the Black–Scholes formula with volatility σ_0 .

Finally note that, unlike ‘traditional’ stochastic volatility models, the interest rate r appears in the ‘volatility’ SDE and consequently in the fundamental PDE (10). This has the consequence that the BS implied volatility takes on an additional dependence on r that cannot be absorbed into the money variable m above. This nuance is discussed further below.

10 Model asymptotics

Some natural questions are: how does the model behave for large and small λ , recalling that λ sets the time scale for which history matters? In addition: how does the implied volatility surface behave for large and small T , where T is the time to an option maturity? Asymptotics help clarify the mathematical behavior of the model, which generally requires a numerical solution. Asymptotics may help create relatively faster calibrations, using them (when appropriate) as a substitute for our PDE solvers.

10.1 The implied volatility smile as $T \rightarrow 0$

Option prices equal their payoff function at $T = 0$. What is interesting is that, for any diffusion model (like ours), there is a nontrivial limit as $T \rightarrow 0$ for the Black–Scholes (BS) implied volatility, call it Σ . Recall the Black–Scholes formula for a call option on a non-dividend-paying stock:

$$c_{BS}(T, S_0, K, \sigma) = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2),$$

with well-known $d_{1,2}$. Consider any (time-homogeneous Markov) n -dimensional diffusion model for a call option price. The model has stochastic factors $x = (S, Y)$ with x an n -vector and Y an $(n-1)$ -vector. (In our anchoring model, $n = 2$ and $Y = A$ or, $Y = \xi$ take your pick.) Let $C(T, S_0, Y_0; K)$ denote that general model option price, at time $t = 0$, for option maturity T and strike K . Then, Σ is defined by equating the model price to the BS price:

$$C(T, S_0, Y_0; K) = c_{BS}(T, S_0, K, \Sigma). \quad (11)$$

Of course, for this to work, we must allow $\Sigma = \Sigma(T, K, S_0, Y_0)$. Remarkably, the complicated nonlinear inversion associated with (11), in general diffusion models, admits an (asymptotic) expansion of the form

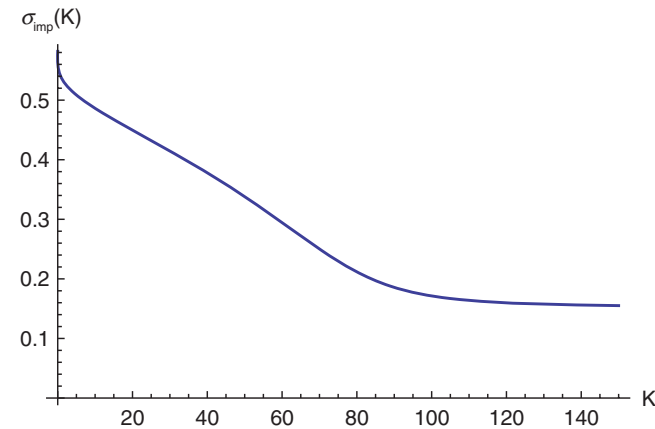
$$\Sigma = \Sigma^{(0)} + T\Sigma^{(1)} + T^2\Sigma^{(2)} + \dots, \quad \text{as } T \rightarrow 0. \quad (12)$$

So, for the limiting expiration smile, we have $\Sigma^{(0)}(K, S_0, Y_0)$ in general and $\Sigma^{(0)}(K, S_0, A_0)$ for our particular model here. Notice that in Table 1, we show entries for $T = 0$. They were obtained from the following formula:

$$\Sigma^{(0)}(K, S_0, A_0) = \frac{\log S_0/K}{\int_{K/A_0}^{S_0/A_0} \frac{dx}{x\sigma(x)}}, \quad (13)$$

where we recall that $\sigma(x)$ is our sigmoidal function. At $K = S_0$, an application of l’Hopital’s rule is implied, which yields the asymptotic at-the-money implied volatility $\sigma(S_0/A_0)$. [For Tables 1 and 2, $S_0 = A_0$ and $\sigma(1) \approx 0.171342$.] If $\sigma(x) = \sigma_0$, a constant,

Figure 8: Asymptotic $T \rightarrow 0$ implied volatility vs. strike. The figure shows a plot of the formula in equation (13) for the parameter set of Table 2.



then (3) (with $\mu = r$) is the Black–Scholes SDE. Reassuringly, in that case, (13) correctly regenerates $\Sigma^{(0)} = \sigma_0$.

Figure 8 shows a plot of the formula in (13) for the parameter set of Table 2. In the sigmoidal function, $a = 0.60$, which is the maximum ‘local’ volatility; this is also seen in the figure to be the maximum implied volatility, as one would expect. Similarly, $b = 0.15$ in the minimum local and implied volatility.

At-the-money relationship. In practice, as T grows small, liquidity in options becomes concentrated near-the-money. By taking successive K -derivatives of equation (13), and then setting $K = S_0$, one can derive the following at-the-money relations. To this end, abbreviating $\Sigma(K) \equiv \Sigma^{(0)}(K, S_0, A_0)$ and with $\xi_0 = S_0/A_0$, we find

$$\Sigma(S_0) = \sigma(\xi_0),$$

$$S_0 \Sigma'(S_0) = \frac{1}{2} \xi_0 \sigma'(\xi_0),$$

$$S_0^2 \Sigma''(S_0) = \frac{1}{3} \xi_0^2 \sigma''(\xi_0) - \frac{3}{2} \xi_0 \sigma'(\xi_0) - \frac{7}{6} \xi_0^2 \frac{[\sigma'(\xi_0)]^2}{\sigma(\xi_0)}.$$

Derivation. Briefly, here is how (13) is obtained. We start with call option values in general n -dimensional models:

$$C(T, S_0, Y_0; K) = e^{-rT} \int \max(S_T - K, 0) q(T, S_0, Y_0; S_T) dS_T,$$

where q is the transition density for a particle starting at $x_0 = (S_0, Y_0)$ to arrive at S_T with any value of Y_T . Now, the asymptotic smile is obtained by taking two K -derivatives of both sides of (11) and comparing them as $T \rightarrow 0$. First suppose the underlying n -factor diffusion has variance–covariance matrix $a(x) = \{a_{ij}(x)\}$. Denote the inverse of this matrix by $g(x)$, so $g(x) = a^{-1}(x)$.

First, two K derivatives of the right-hand-side. Since the BS formula is based on the lognormal density, one has

$$\frac{\partial^2 c_{BS}}{\partial K^2}(T, S_0, K, \Sigma) \approx \exp \left\{ -\frac{(\log S_0/K)^2}{2(\Sigma^{(0)})^2 T} \right\}. \quad (14)$$

(Note: all “ $\approx$ ” denote leading behaviors as $T \rightarrow 0$, up to ignorable factors.)

Now the left-hand-side. From Breeden–Litzenberger’s (1978) relation and a classic argument due to Varadhan (1967):

$$\frac{\partial^2 C}{\partial K^2}(T, S_0, Y_0; K) = e^{-rT} q(T, S_0, Y_0; K) \approx \exp \left\{ -\frac{d^2(S_0, Y_0; K)}{2T} \right\}, \quad (15)$$

where

$$d^2(S_0, Y_0; K) \equiv \min_{\substack{x(0)=(S_0, Y_0) \\ x(1)=(K, free)}} \int_0^1 g_{ij}(x(s)) \dot{x}^i(s) \dot{x}^j(s) ds.$$

In the integral shown for d^2 , the ‘min’ is a functional minimization over all parameterized path functions $\{x(s): 0 \leq s \leq 1\}$ subject to the endpoint conditions shown. The scalars x^i are simply the components of x , $\dot{x}^i(s) = dx^i/ds$, and there is an implied summation over repeated indices. With that notation, $d(S_0, Y_0; K)$ has a fascinating interpretation. It is the minimum geodesic distance in the (S, Y) space from (S_0, Y_0) to the (hyperplane) $S = K$. The geodesics are those found by treating $g = a^{-1}$ as a metric and doing deterministic (that is, non-stochastic) path calculations. Comparing the right-hand sides of (14) and (15) yields a general result for the asymptotic smile:

$$\Sigma^{(0)}(K, S_0, Y_0) = \left| \frac{\log S_0/K}{d(S_0, Y_0; K)} \right|. \quad (16)$$

Thus it comes down to calculating geodesic distance functions $d(x, A)$, where x is again an n -dimensional point and A is *any set* in the factor variable space that does not contain x . There are a variety of ways to do this. One of the most straightforward ways is to solve the *Eikonal* equation,⁷ namely

$$a_{ij}(x) \frac{\partial d}{\partial x^i} \frac{\partial d}{\partial x^j} = 1,$$

subject to the boundary condition that $d(x, A)$ vanishes when x touches A . For example, start with $n = 1$ and local volatility evolutions $dS_t = \tilde{\sigma}(S_t)S_t dW_t$. A drift may be present, but it plays no role in the limiting smile, so we ignore it. Then, $a(S) = \tilde{\sigma}(S)^2 S^2$ and $d = d(S; K)$. Thus the Eikonal problem is easy: solve $S^2 \tilde{\sigma}(S)^2 (d_S)^2 = 1$, subject to $d(S = K) = 0$. This has the solution

$$d(S; K) = \int_K^S \frac{du}{u \tilde{\sigma}(u)}. \quad (17)$$

Now our problem is a two-factor problem, $dS_t = \sigma(S_t/A_t)S_t dW_t$, but the second factor A_t has an SDE with a drift but no Brownian term. Given that degeneracy, one would guess that local volatility formula (17) applies with $\tilde{\sigma}(S) = \sigma(S/A_0)$. We recall that $\sigma(\cdot)$ is the sigmoidal function. That is,

$$d(S_0, A_0; K) = \int_K^{S_0} \frac{du}{u \sigma(u/A_0)} = \int_{K/A_0}^{S_0/A_0} \frac{dx}{x \sigma(x)}.$$

Applying (16), this last relation yields our formula (13). Our guess can be justified by setting up the full problem in two-dimensional (S, ξ) space, which is governed by a bivariate SDE with two (perfectly correlated) Brownian noise terms. Then, one shows that the putative solution for the distance function (but written in terms of S and ξ) satisfies the corresponding two-dimensional Eikonal equation. It is tedious, but straightforward, and a good exercise for the ambitious reader.

Further reading. The geodesic connection is developed in Varadhan (1967). See Lewis (2007) for a more elaborate but heuristic derivation of (16) and further references. See also the chapter on ‘Advanced smile asymptotics’ in Lewis (2014, forthcoming).

10.2 The implied volatility smile as $T \rightarrow \infty$

The method is to develop the large- T behavior of the Fourier solution representation (8) and especially the behavior of the ‘fundamental transform’ $H(T, \xi; z)$ in this limit. This method was developed in detail in Lewis (2000), so we will be brief.

In a nutshell, in general models, the implied volatility surface flattens at large maturity T . The flattening is to a common value independent of the moneyness and initial factors. In our case, the initial factors are (S_0, ξ_0) . One has to be careful about what is held fixed as $T \rightarrow \infty$. The natural thing is to fix the moneyness $m = \log \{S_0 e^{-qT}/K e^{-rT}\}$,

Figure 9: ‘Large T asymptotics’ is an eigenvalue problem. As explained in the text, the asymptotic implied volatility as $T \rightarrow \infty$ is associated with an eigenvalue problem $\mathcal{L}u_0(\xi) = \Lambda_0 u_0(\xi)$, where $\Lambda_0 = \Lambda_0(y^*)$. For the parameter set of Table 2, but with $r = 0$, the figure shows $\Lambda_0(y)$ and the associated eigenfunction at $y = y^*$, the maximum point.

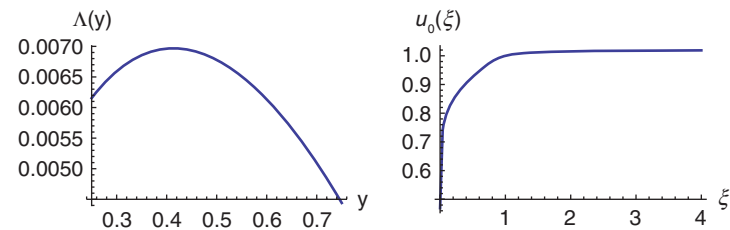


Table 3: Black–Scholes implied volatilities (%): large T and $T \rightarrow \infty$

T	$r = 0$	$r = 0.02$	$r = 0.04$	$r = 0.06$	$r = 0.08$	$r = 0.10$
1000	23.58	21.61	19.94	18.56	17.50	16.71
∞	23.61	21.68	20.00	18.62	17.54	16.74

Table 4: Option prices corresponding to the $T = 1000$ entry in Table 3

$r = 0$	$r = 0.02$	$r = 0.04$	$r = 0.06$	$r = 0.08$	$r = 0.10$
99.981	99.937	99.838	99.664	99.434	99.176

Parameter set: $a = 0.6$, $b = 0.15$, $c = 10$, $d = -0.3$, $\lambda = 0.5$. For each r , the strike was set to the forward at-the-money value: $K = S_0 e^{rT}$.

and that is assumed throughout our discussion.⁸ With that, one can develop the simple relation for the ultimate implied variance

$$V_{\infty}^{imp} = \lim_{T \rightarrow \infty} V_{\infty}^{imp}(T, m, \xi_0) = 8 \Lambda_0(y^*). \quad (18)$$

Then, $\sigma_{\infty}^{imp} = \sqrt{V_{\infty}^{imp}}$. In (18), $\Lambda_0(y)$ is the principal (i.e., smallest) eigenvalue associated with the (negative of the) spatial differential operator of (10), fixing $z = iy$ ($0 < y < 1$). This leads to the eigenvalue problem $\mathcal{L}u_0(\xi) = \Lambda_0 u_0(\xi)$, suppressing the dependence on the fixed y . The function space for u_0 is: smooth enough functions with certain boundary behaviors. Then, y^* is a stationary point; i.e., $\Lambda_0(y^*)$ is the largest of such principal eigenvalues along the line segment $y \in (0, 1)$. For the parameter set of Table 2, but with $r = 0$, Figure 9 shows $\Lambda_0(y)$ and the associated eigenfunction at $y = y^*$, the maximum point.

From (18), and the parameter set of Table 2, except for various interest rates r , Table 3 shows computed values of σ_{∞}^{imp} . To check the results (see Tables 3 and 4), the full time-dependent problem has also been solved with $T = 1000$. This was done by using the Fourier transform method, solving (10) for $H(T, \xi; z)$ in Mathematica with NDSolve. For the finite- T solver call, no boundary conditions were taken at $\xi = 0$ and a zero ξ -derivative was taken at a cutoff $\xi_{max} = 2$. One can see that the finite maturity results are consistent with asymptotic values.

If one carried out this same exercise in, say, the Heston (1993) stochastic volatility model, all the values in Table 3 for σ_{∞}^{imp} would be the same. This is because the ultimate implied volatility flattens to a value that depends on the parameters in the volatility process, which does not contain r . Here, as noted earlier, such r -dependence is present.

Further details of the method, as applied to this particular problem, are rather involved and may be reported elsewhere.

10.3 Model prices as $\lambda \rightarrow 0$

Using subscripts in S and τ for derivatives, and $\tau = T - t$, write equation (5) as $\mathcal{L} V = -\lambda(S - A)V_A$. Here, for any smooth function $f(\tau, S, A)$, we define

$$\mathcal{L} f(\tau, S, A) \equiv -f_\tau + \frac{1}{2}\sigma^2\left(\frac{S}{A}\right)S^2 f_{SS} + rSf_S - rf.$$

By substitution, it is clear that there is a formal power series solution in λ ; that is,

$$V = V^{(0)} + \lambda V^{(1)} + \lambda^2 V^{(2)} + \dots,$$

where the $V^{(n)}$ are recursively determined by $\mathcal{L} V^{(n)} = -(S - A)V_A^{(n-1)}$. Now $V^{(0)}$ is the solution to

$$V_\tau^{(0)} = \frac{1}{2}\sigma^2\left(\frac{S}{A}\right)S^2 V_{SS}^{(0)} + rSV_S^{(0)} - rV^{(0)}, \quad (19)$$

subject to appropriate initial and boundary conditions for the particular payoff being considered. Suppose that we found $V^{(0)}$. Since it will satisfy the right initial and boundary conditions for the full problem (that is, the problem for V), that means all the subsequent $V^{(n)} (n \geq 1)$ should be solved for with *zero* initial and boundary conditions. This is well-posed because each such subsequent problem has a driving term.

Now let us restrict ourselves to the problems for vanilla puts and calls with strike K , writing $V = V(\tau, S, A; K)$. Take the put; we can write $V^{(0)}(0, S, A; K) = (K - S)^+ = A(k - \xi)^+$, using $\xi = S/A$, $k = K/A$. This suggests we should look for a solution to (19) in the form $V^{(0)}(\tau, S, A; K) = Au^{(0)}(\tau, \xi; k)$. Indeed, $u^{(0)}$ solves

$$u_\tau^{(0)} - \left\{ \frac{1}{2}\sigma^2(\xi)\xi^2 u_{\xi\xi}^{(0)} + r\xi u_\xi^{(0)} - ru^{(0)} \right\} = 0 \quad (20)$$

subject to $u^{(0)}(0, \xi; k) = (k - \xi)^+$, and similarly for the call. This is readily solved numerically – certainly much easier than the previous PDEs with two spatial factors. Then, at the next order, $V^{(1)}(\tau, S, A; K) = Au^{(1)}(\tau, \xi; k)$. Now $u^{(1)}$ solves

$$u_\tau^{(1)} - \left\{ \frac{1}{2}\sigma^2(\xi)\xi^2 u_{\xi\xi}^{(1)} + r\xi u_\xi^{(1)} - ru^{(1)} \right\} = f(\tau, \xi; k), \quad (21)$$

where

$$f(\tau, \xi; k) = (1 - \xi)\left(\xi u_\xi^{(0)} + k u_k^{(0)} - u^{(0)}\right),$$

and with $u^{(1)}(0, \xi; k)$. The same equation applies to all orders, except that the left-hand side uses $u^{(n)}$ and the right-hand side uses $u^{(n-1)}$. We have carried this through for the first two orders in Mathematica, and show the following numerical results. Consider the Table 2 entries at $T = 1$ and $T = 3$. We show the results for $V^{(0)}$, $\lambda V^{(1)}$ and their sum in Table 5. Comparing with Table 2, the expansion results are very good at $T = 1$. But they suggest that at least the $O(\lambda^2)$ correction is necessary for a good result at $T = 3$. In general, the controlling parameter here is the dimensionless λT .

Table 5: European-style put option prices: λ expansion

T = 1	K = 60	K = 80	K = 100	K = 120	K = 140
$V^{(0)}$	0.230	0.986	4.717	16.255	33.545
$\lambda V^{(1)}$	−0.066	−0.155	−0.043	0.047	0.002
Sum	0.164	0.831	4.674	16.302	33.547
T = 3	K = 60	K = 80	K = 100	K = 120	K = 140
$V^{(0)}$	1.458	2.975	6.249	13.615	25.334
$\lambda V^{(1)}$	−0.514	−0.766	−0.371	0.054	0.030
Sum	0.945	2.209	5.878	13.669	25.365

Large λ . Finally, to complete the asymptotics, we note that as $\lambda \rightarrow \infty$, this drives $A_t \rightarrow S_t$. So $\sigma(S_t/A_t) \rightarrow \sigma(1)$ and the model prices should converge to the Black–Scholes’ model prices with constant $\sigma = \sigma(1)$.

11 Calibration challenges and conclusions

In principle, given a parameterized version of $\sigma(\xi)$, one can estimate the parameters by fits to SPX option chains. Neither of the two Mathematica implementations here are efficient enough for this. However, we can refer the reader to Foschi and Pascucci (2009), who have managed to achieve such a calibration in the closely related class of models due to Hobson and Rogers (1998).

In general, diffusion models have a very hard time fitting short-dated SPX options. That is because the implied volatility smiles for broad-based indexes such as SPX tend to become increasingly “V-shaped” as maturity approaches. That behavior is strongly suggestive of price jump processes. Since all ‘complete models with stochastic volatility’ are diffusions, calibrations will likely suggest that these classes of models are best embedded as components in (larger) jump diffusions. This will, of course, break the complete-markets property, which is generally contra-indicated anyway.

Acknowledgments

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ENDNOTES

1. Our model is 'complete with stochastic volatility,' in the sense of Hobson and Rogers (1998), and has similar features.
2. We generally denote partial derivatives with a subscript, except where clarity suggests a more explicit notation.
3. Not every choice for $\sigma(x)$ admits a stationary density, and a stationary density is not always needed. When $\sigma(0) < \infty$, as is true for our sigmoidal parameterization, it is easy to see that

$$2 \int_1^\xi \frac{\mu + \lambda - \lambda s}{s\sigma(s)^2} ds \sim \frac{2(\mu + \lambda)}{\sigma^2(0)} \log \xi, \text{ as } \xi \rightarrow 0.$$

This leads to the ξ -stationarity condition: $\mu + \lambda > \sigma(0)^2/2$. When $\sigma(0) < \infty$, but the stationary condition is violated, ξ -mass simply accumulates closer and closer to the natural boundary $\xi = 0$ (similar to GBM). For some applications, say stocks with bankruptcy risk, unbounded $\sigma(\cdot)$ may be desirable. Then, $P_\infty(\xi)$ develops a Dirac mass at $\xi = 0$ and the existence of a stationary density in ξ becomes irrelevant. Finally, nothing much interesting happens for large ξ : for integrability there, it is sufficient that $\lambda > 0$ and $\sigma(\infty) < \infty$. The latter condition is uncontroversial for most equities, as this is the regime of large S relative to history.

4. The standard NDSolve algorithm is a 'method of lines.' Basically: apply a spatial PDE discretization, which converts the PDE into an ODE system. The latter is then solved by an adaptive time-stepping routine. There are many details.
5. Such processes are sometimes termed *Markov Additive Processes*, or MAPs, in the applied probability literature.
6. Those are very weak conditions. Our anchoring model, with a sigmoidal function bounded away from 0 and infinity, should be both norm-preserving and martingale-preserving – although we do not prove it.
7. Sometimes called the Hamilton–Jacobi equation.
8. Suppose S_0 is fixed and we increase T . Note that fixing the moneyness is not the same as fixing the strike K , except when $q = r$.

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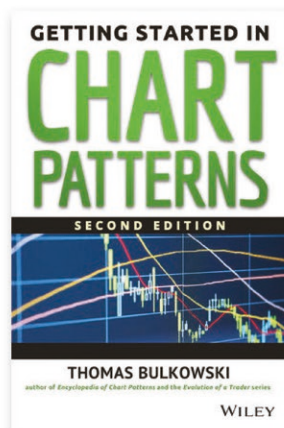
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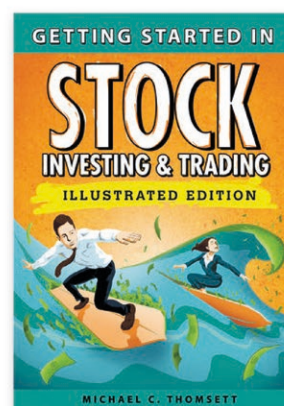


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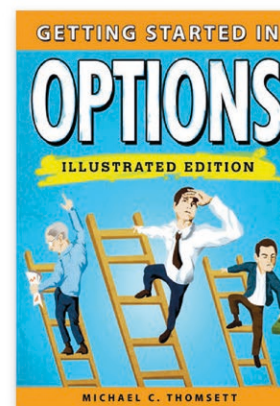
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